

Homework Assignment 2 (Rewrite)

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1. (1.A.8) Suppose $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable. Prove that

$$\int_a^b f = \lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{j=1}^n f \left(a + \frac{j(b-a)}{n} \right). \quad (1)$$

Proof. Define a partition $P_n = \left\{ a, a + \frac{b-a}{n}, \dots, a + \frac{(n-1)(b-a)}{n}, b \right\}$ for each $n \in \mathbb{Z}^+$. Let $\varepsilon > 0$ and choose P such that $L(f, P, [a, b]) > \int_a^b f - \frac{\varepsilon}{2}$.

Then

$$\int_a^b f - \frac{\varepsilon}{2} < L(f, P, [a, b]) \leq L(f, P \cup P_n, [a, b]). \quad (2)$$

We can choose m large enough (needs proof) so that

$$\begin{aligned} L(f, P \cup P_m, [a, b]) - L(f, P_m, [a, b]) &< \frac{\varepsilon}{2} \\ \implies -\frac{\varepsilon}{2} &< L(f, P, [a, b]) - L(f, P \cup P_m, [a, b]). \end{aligned}$$

Adding this to (2), we get

$$\int_a^b f - \varepsilon < L(f, P_m, [a, b]).$$

An analogous argument gives us an m such that

$$\int_a^b f + \varepsilon > U(f, P_m, [a, b]).$$

Finally, we observe that since

$$\inf_{\left[a + \frac{(j-1)(b-a)}{n}, a + \frac{j(b-a)}{n} \right]} f \leq f \left(a + \frac{j(b-a)}{n} \right) \leq \sup_{\left[a + \frac{(j-1)(b-a)}{n}, a + \frac{j(b-a)}{n} \right]} f,$$

$$L(f, P_n, [a, b]) \leq \frac{b-a}{n} \sum_{j=1}^n f \left(a + \frac{j(b-a)}{n} \right) \leq U(f, P_n, [a, b]),$$

and thus since we showed that the lower and upper sums over P_n create sequences which limit to $\int_a^b f$, the middle sum above is squeezed so that (1) is satisfied. $\square$

2. (1.A.12) Suppose $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable. Prove that $|f|$ is Riemann integrable and that

$$\left| \int_a^b f \right| \leq \int_a^b |f|.$$

Proof. Given a partition P of $[a, b]$, since $x \leq |x|$ for all $x \in \mathbb{R}$, and thus $\inf_A f < \inf_A |f|$ and $\sup_A |f| \leq \sup_A f$,

$$L(f, P, [a, b]) \leq L(|f|, P, [a, b]) \leq U(|f|, P, [a, b]) \leq U(f, P, [a, b]). \quad (3)$$

Since f is Riemann integrable, for any $\varepsilon > 0$, we can find a partition P_ε such that

$$U(f, P_\varepsilon, [a, b]) - L(f, P_\varepsilon, [a, b]) < \varepsilon,$$

so by (3), we have □

3. (1.B.1) Define $f : [0, 1] \rightarrow \mathbb{R}$ as follows:

$$f(a) = \begin{cases} 0 & \text{if } a \text{ is irrational,} \\ \frac{1}{n} & \text{if } a \text{ is rational and } n \text{ is the smallest positive integer} \\ & \text{such that } a = \frac{m}{n} \text{ for some integer } m. \end{cases}$$

Show that f is Riemann integrable and compute $\int_0^1 f$.

Proof. Let $P = \{0 = p_0, p_1, \dots, p_n = 1\}$ be a partition of $[0, 1]$. Since there is an irrational number within any interval $[p_{i-1}, p_i]$, $\inf_{[p_{i-1}, p_i]} f = 0$, and thus $L(f, P, [0, 1]) = 0$. We proceed to show that for any $\varepsilon > 0$, there exists a partition P such that

$$U(f, P, [0, 1]) - L(f, P, [0, 1]) = U(f, P, [0, 1]) < \varepsilon.$$

Let $\varepsilon > 0$, and let p be a prime such that $1/p < \varepsilon$. Then consider the partition $P = \{0, \frac{1}{p}, \frac{2}{p}, \dots, \frac{p-1}{p}, 1\}$. □

4. (1.B.4) Give an example of bounded functions $f, g : [0, 1] \rightarrow \mathbb{R}$ such that

$$L(f, [0, 1]) + L(g, [0, 1]) < L(f + g, [0, 1])$$

and

$$U(f + g, [0, 1]) < U(f, [0, 1]) + U(g, [0, 1]).$$

Proof. Define

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ -1 & \text{if } x \notin \mathbb{Q} \end{cases}$$

and

$$g(x) = \begin{cases} -1 & \text{if } x \in \mathbb{Q} \\ 1 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Then $(f + g)(x) = 0$ for all $x \in [0, 1]$, so

$$L(f, [0, 1]) + L(g, [0, 1]) = -1 - 1 = -2 < 0 = L(f + g, [0, 1])$$

and

$$U(f, [0, 1]) + U(g, [0, 1]) = 1 + 1 = 2 > 0 = U(f + g, [0, 1]).$$

□

5. (2.A.1) Prove that if A and B are subsets of $\mathbb{R}$ and $|B| = 0$, then $|A \cup B| = |A|$.

Proof. By countable subadditivity,

$$|A \cup B| \leq |A| + |B| = |A| + 0 = |A|.$$

By monotonicity, since $A \subset A \cup B$,

$$|A| \leq |A \cup B|.$$

Therefore, $|A \cup B| = |A|$.

□

6. (2.A.8) Prove that if $A \subset \mathbb{R}$ and $t > 0$, then $|A| = |A \cap (-t, t)| + |A \cap (\mathbb{R} \setminus (-t, t))|$.

Proof.

□